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ORIGINAL RESEARCH

Classifying legal interpretations using large language models

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Abstract
In the civil law tradition, legal arguments are used to justify the outcomes of judicial decision-making. These arguments are formed relying on a canon of interpretation techniques (e.g. textual or teleological interpretation). We study the identifiability of interpretation techniques as they are employed by the European Court of Human Rights (ECtHR) from a computational law perspective using a unique dataset. We show how Large Language Models (LLMs) can be utilized to classify legal interpretations, and we compare their performance. We evaluate proprietary and opensource models using methods such as few-shot and zero-shot chain-of-thought prompting combined with self-consistency. Our results imply that feature-extraction using LLMs leads to robust outcomes while allowing for greater resource- and timeefficiency compared to human annotation. Furthermore, our results imply that LLMs can play a larger role in the extraction of more complex features that are of particular relevance from a legal perspective.

Keywords Legal interpretations · Computational Law · European Court of Human Rights · Large Language Models

1 Introduction

Law is fundamentally about making legal arguments. Particularly in the civil law tradition, legal arguments are commonly constructed by employing various interpretation techniques (e.g. textual interpretation) on the legal sources (e.g., a statutory

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legal rule) (Walton et al 2018; Greenawalt 2012). Deriving arguments from legal sources through the application of interpretation techniques is an important part of legal reasoning carried out by the courts in countries following the civil law tradition. In fact, the use of interpretation techniques is a quality indicator of good legal reasoning by the courts (with a focus on the international judiciary (Von Bogdandy and Venzke 2014)). Furthermore, law students are trained both in the identification and construction of legal arguments employing the various techniques of interpretation (Huhn 2000).

Despite their importance for both education and practice, the use of interpretation techniques is rarely studied from the perspective of computational law. This is certainly due to the fact that it is hard to provide clear definitions even for the canonical techniques of interpretation (textual, systematic, historical and teleological interpretation) (for a proposal see Walton et al (2018)). Given the ambiguity of the definitions of the individual interpretation techniques, classification tasks carried out by humans remain somewhat subjective.

This paper focuses on the judicial interpretation of legal rules contained in the European Convention on Human Rights (ECHR). Specifically, we look at the use of the techniques of interpretation by the European Court of Human Rights (ECtHR). The ECtHR is one of the most influential human rights courts worldwide. It currently has 46 Member States, and its case-law is relevant for more than 700 m people.[1] The ECtHR is composed of judges from all 46 Member States; the judges have different backgrounds (some of them are career judges with a long-standing experience in the practice of law, others are law professors), some are trained in the civil law, others in the common law tradition.

Research into trends regarding the ECtHR's usage of certain interpretation techniques relies on having the appropriate data. To the best of our knowledge, data on the use of interpretation techniques by the ECtHR is scarce. The main reason for the data scarcity on techniques of legal interpretation is that the annotation of such data is not an easy task, and automating such a process is far from trivial. An appropriate collection process would require parsing through lengthy judgments while meticulously analyzing the text for the presence of certain interpretative arguments. The training of classical machine learning methods to solve this issue is an option, albeit a difficult one. In order to achieve sufficient accuracy, large amounts of training data are necessary. If one had such data, one could employ standard classification techniques.

To circumvent the lack of large sets of training data, we suggest the usage of Large Language Models (LLMs) for the classification of legal interpretations. The release of the Generative Pre-trained Transformer 3 (GPT-3) (Brown et al 2020) thrusted LLMs and the few-shot learning setting into prominence. Since then major developments in the field have been made at an increasingly higher pace. Naveed et al (2023) show that the number of models released in the past years has been strongly increasing, totaling to 29 for the year 2023. They also show that the vast

[1] https://www.coe.int/en/web/portal.

majority of models released in 2023 were open-source models, a shift from previous years.

In this paper we evaluate several proprietary and open-weights models for classifying techniques of legal interpretation on text excerpts obtained from judgments by the ECtHR. We aim to classify three legal interpretation techniques - textual, systematic, and teleological. In Sect. 2, we provide a brief overview of legal interpretation techniques and their use by the ECtHR. In Sect. 3, we discuss related work. Using custom tailored prompts, in Sect. 4.2 we use this data to perform zero-shot and few-shot learning on our selection of models. In Sect. 4.2.1, we use chain-of-thought prompting as introduced by Wei et al (2023) in combination with the zero-shot reasoning prompting strategies suggested in Kojima et al (2023). We draw comparisons between this approach and the results obtained through standard zero- and few-shot learning strategies. Furthermore, following Wang et al (2023), in Sect. 4.3 we implement the notion of self-consistency which is reported to lead to improvements in performance for chain-of-thought-prompting. In Sect. 5 we show that these two strategies both lead to improvements in performance in certain settings. Furthermore, we show these methods outperform a baseline BERT-based (Devlin et al 2019) classifier, LEGAL-BERT (Chalkidis et al 2020), pretrained on ECHR cases. Section 5.2 investigates similarities in the models' reasoning language compared to human-written reasoning. Our research is the first step towards an automated process for the collection of interpretations data from large-scale legal documents.

2 Overview of legal interpretation techniques and their use by the ECtHR

According to Art. 45(1) ECHR, the ECtHR must give reasons for its judgments and decisions. In the process of formulating legal arguments, the ECtHR often makes use of the canon of legal interpretation techniques. To be more precise, the ECtHR refers to these techniques when establishing the meaning of a Convention norm (the goal of legal interpretation being to "assign meaning to texts and other statements for the purposes of establishing rights, obligations, and other consequences relevant in a legal text" (Herdegen 2013)). In order to identify the meaning of a legal text, judges employ a set of interpretation techniques. In international law, the canonical techniques of legal interpretation are textual, systematic, historical and teleological interpretation (Hollis 2019). These techniques are laid out in Articles 31-33 of the Vienna Convention on the Law of Treaties.[2] Following these rules, the ECtHR also regularly applies the canonical techniques in its case-law (Dothan 2018). In addition, the ECtHR developed further techniques of interpretation (e.g. evolutionary interpretation, margin of appreciation) (Dothan 2018). However, in practice, the ECtHR relies most frequently on three of the canonical techniques of interpretation: textual, systematic and teleological interpretation (Altwicker et al 2024). Given

[2] https://legal.un.org/ilc/texts/instruments/english/commentaries/1_1_1966.pdf.

that historical interpretation and the court-specific techniques are rarely used by the ECtHR, our work focuses on the small subset of the three most pervasive techniques.

Textual interpretation refers to understanding the ordinary meaning of a term. This method utilizes the dictionary sense to establish the meaning (Slocum and Wong 2021); from a linguistic perspective (Hutton 2020). In specific fields like science, law, or technology, the subject-specific meanings are critical. Sometimes, therefore, the literature distinguishes arguments from ordinary meaning and arguments from technical meaning (Walton et al 2018). Usually, textual interpretation relies on the ordinary meaning at the time of conclusion of the treaty (Dörr 2018). In the context of the European Convention on Human Rights (ECHR), textual interpretation often serves as a starting point only and sometimes as a check on other techniques of interpretation (see Polgári (2021)). An example for the use of the textual interpretation technique from the case-law of the ECtHR is this: "[A]s a matter of textual interpretation, the wording of Article 5 § 4, clearly suggests that the court must have the power to order a release if it finds the detention unlawful" (Case of Buishvili v. the Czech Republic, judgment of 25 October 2012, app. no. 30241/11, § 39).[3]

Systematic interpretation looks at the context and relationships between different terms or rules (Walton et al 2018); on the theoretical prerequisites see Padjen (2020). It involves considering the entire treaty, including preambles and annexes (Dörr 2018). This technique of interpretation is based on the hypothesis that legal rules are interconnected or part of a normative "pattern" (Ammann 2020); the use of this technique aims to make a body of legal rules consistent (Feteris and Feteris 2017). The ECtHR has developed a rich case-law making use of systematic interpretation, often indicated by the phrase of interpreting a Convention right "in the light of" another Convention right. This is an example: "The two sentences of Article 2 of Protocol No. 1 must be interpreted not only in the light of each other but also, in particular, of Articles 8, 9 and 10 of the Convention..." (Case of Folgerø v. Norway, judgment of 29 June 2007, app. no. 15472/02, § 84). Sometimes, the ECtHR uses systematic interpretation by linking Convention rights to (higher-level) Convention principles (see, e.g., the principle of the "rule of law" in the Case of Al-Dulimi and Montana Management Inc. v. Switzerland, app. no. 5809/08, judgment of 21 June 2016, § 126).

Teleological interpretation focuses on the purpose and objectives that the lawmaker intended to achieve with a rule (Walton et al 2018). The use of the teleological interpretation technique requires taking into account the goals and the legal interests the legal rule is to realize or protect (Ammann 2020). Teleological interpretation is commonly considered as a technique that allows the judges most flexibility in the construction of legal meaning (for the ECtHR see Dothan (2014)). An example from the case-law in which teleological interpretation was employed is this: "Observing that religious communities traditionally exist in the form of organised structures, the Court has repeatedly found that the autonomous existence of religious communities is indispensable for pluralism in a democratic society and is, thus, an

[3] https://hudoc.echr.coe.int/.

issue at the very heart of the protection which Article 9 affords" (Case of Löffelmann v. Austria, judgment of 12 March 2009, app. no., 42967/98, § 47).

Classifying interpretation techniques in ECtHR judgments presents several challenges. First, as already mentioned, there are no clear, universal definitions on the interpretation techniques, making the classification task subjective to some degree. Second, interpretation methods often overlap (e.g. teleological and systematic interpretation), making it difficult to distinguish between them in practice. Third, the ECtHR's judgments are often highly contextual, influenced by specific legal, cultural, and political circumstances, requiring an understanding of the broader context. Fourth, judges frequently do not explicitly label the interpretative methods they employ when forming legal arguments about the Convention rights, necessitating inference based on broader patterns of reasoning. These factors make classifying interpretation techniques in ECtHR judgments a complex task for human annotators.

3 Related work

Artificial intelligence in the context of law and legal research has been much discussed (Chalkidis and Kampas 2019). For example, Nguyen et al (2018) use recurrent neural networks to identify requisite and effectuation parts in legal texts, Chalkidis et al (2018) use hierarchical recurrent neural networks for detecting contractual obligations and prohibition from text, Merchant and Pande (2018) use latent semantic analysis for legal text summarization, while Chalkidis et al (2020) develop LEGAL-BERT, an adaption of BERT (Devlin et al 2019) obtained by performing pre-training on legal domain-specific data.

However, empirical research on legal interpretation techniques is limited. Zurek and Araszkiewicz (2013) provide a mathematical model for teleological interpretations in the context of the provision of statutory law. Habernal et al (2023) annotated legal arguments, including interpretation techniques, in 373 ECtHR judgments. However, the three interpretations at the focus of this paper only make up a combined 0.43% of the annotated dataset (a total of 55 observations). Furthermore, the majority of the models evaluated in their work fail to make a single correct prediction for any of the interpretation techniques. The only exception was their variation of RoBERTa (Liu et al 2019) pretrained on legal data which shows some predictive power for teleological interpretations. Their work is further proof of the difficulty of collecting legal interpretations data, as well as an opportunity to evaluate methods requiring less training data.

The advent of LLMs has had a profound impact on the field of law. Much research has been done in areas where the proponents of LLMs believe such models can assist, or even replace, lawyers. Choi et al (2023) report ChatGPT successfully passes several law school exams, while Katz et al (2023) report GPT-4 achieves a well-above passing grade on the entirety of the Uniform Bar Examination. Models such as GPT-3 showed capability in understanding consumer contracts, as well as in performing legal rule classification (Kolt 2022; Liga and Robaldo 2023). Savelka (2023) evaluates the models' capabilities in performing zero-shot semantic annotation on text excerpts from a variety of legal documents.

LLMs have shown their reasoning capabilities on a variety of tasks (Wei et al 2023). Nay et al (2023) evaluate reasoning capabilities of several models when it comes to applying tax law, while Trozze et al (2023) study their reasoning and legal drafting capabilities on security cases involving cryptocurrencies.

Despite their reasoning abilities, LLMs have been known to hallucinate (Ji et al 2023; Ye and Durrett 2022), that is produce seemingly convincing text containing completely false information. The discussion of interpretability and security of LLMs is ongoing (Bender et al 2021; Shen et al 2023; Yao et al 2024; Saha et al 2023) and will remain important going forward. To circumvent these issues, as well as boost performance, several prompting methods have been researched. Self-consistency introduced by Wang et al (2023) has shown to improve reasoning performance, while Kojima et al (2023) introduce specific prompting strategies to aid reasoning. More specifically to the legal domain, Jiang and Yang (2023) use legal syllogism to construct chain-of-thought prompts, while Yu et al (2022) derive prompts from specific legal reasoning techniques such as IRAC (Issue, Rule, Application, Conclusion). Furthermore, there has been increased standardization in how these models are evaluated on legal tasks through the development of benchmarks such as LegalBench (Guha et al 2023).

While the usage of LLMs for text classification and annotation has been evaluated on a variety of topics, to the best of our knowledge, their use in the classification of legal interpretations has not yet been reported. We deem this work as the first step towards using these models for identifying legal interpretations in large-scale legal documents. Our hope is this paper can further the use of such models in the field of law for automated interpretative technique annotation.

4 Analysis setting

4.1 Data and models

We extract ECtHR judgments of the years from 2006 till 2011 from the HUDOC[4] database focusing on the majority opinion only (i.e. leaving out separate opinions). These judgments were further split into smaller text excerpts. The data used in this paper was randomly sampled from a larger dataset consisting of 1181 observations while ensuring a balanced sample of the 3 interpretations. The original dataset was labeled by 3 law students trained specifically for this task with inter-annotator reliability evaluated using Fleiss' Kappa ($\kappa = 0.93179$) (Fleiss 1971). The labels of the sampled data were then further verified by a law professor. In total, 130 texts were labeled for evaluation purposes. The dataset contains 40 texts with a systematic interpretation, 40 with a textual interpretation, and 40 with a teleological interpretation. In order to evaluate whether models can identify the lack of an interpretative argument, an important feature in order to deploy these models for data collection on large-scale documents, we include 10 texts containing no interpretative

[4] https://hudoc.echr.coe.int/.

arguments. The average length of these texts was 333 characters with a standard deviation of 130 characters.

The evaluated models cover a small range of the most popular LLMs currently available. We consider 4 models in this paper which we deemed most relevant. These include 2 proprietary models, and 2 open-weights models. We include **GPT3.5 Turbo −** by OpenAI (Brown et al 2020), **Llama-2** by Meta AI (Touvron et al 2023), **Gemini Pro** by Google AI (Team et al 2023) and **Mixtral** by MistralAI (Jiang et al 2024). We follow the model querying procedure outlined in (Akter et al 2023) using TogetherAI[5] and LiteLLM.[6]

GPT3.5 Turbo− is a proprietary model developed by OpenAI. The model does not have publicly available information on the training data nor the parameter size. We avoid evaluating GPT-4 and GPT-4 Turbo (OpenAI et al 2023) due to the cost-prohibitive nature of running large-scale evaluations on these models. The current cost, per 1 million tokens, for GPT-4 Turbo is tenfold the price of its contemporaries which makes its usage on any large legal documents unrealistic. Hence, we narrow our focus on GPT−3.5 Turbo. More specifically, we use the *gpt3.5-turbo-preview* − model. The model was accessed through the OpenAI API.

Llama-2 is a family of open-weights LLMs developed by Meta AI. In this paper we narrow our focus on the 70 billion parameters Llama-2 model, the largest one available, henceforth denoted as Llama-2(70b). In addition to being open-weights, the relative small number of parameters of the Llama-2 models compared to its contemporaries has enabled the deployment of local instances of the models. This has lead to a further democratization of the LLM space (Naveed et al 2023). The model was accessed through the API provided by TogetherAI using LiteLLM.

Gemini Pro is a proprietary model developed by Google AI (Team et al 2023). Similarly to the OpenAI models, the training data and the parameter size have not been disclosed. We access the model through the Google VertexAI API. Furthermore, we disable the safety features of the model due to the fact that some of the evaluation data contains material where the court had to adjudicate on matters which Gemini Pro deems sensitive. Google has further announced Gemini Pro 1.5[7] with a context window of up to 1 million tokens. We look forward to evaluating this model as soon as possible since such a large context window is a big step toward our goal of automated interpretations data collection from large-scale documents using LLMs.

Mixtral is an open-weights, sparse mixture-of-experts model which claims to outperform Llama-2(70b) on most benchmarks, as well as offer comparable performance to GPT−3.5 Turbo (Jiang et al 2024). To be more specific, we use the *Mixtral-8x7b-Instruct-v0.1* version of the model. Similarly to Llama-2(70b), we access the model through the TogetherAI API using LiteLLM.

Finally, we compare all these models to a baseline BERT-based model. We use **LEGAL-BERT** by Chalkidis et al (2020) as a baseline. More specifically, we use

[5] https://www.together.ai/.

[6] https://litellm.ai/.

[7] https://storage.googleapis.com/deepmind-media/gemini/gemini_v1_5_report.pdf.

the *ECHR-BERT-BASE* model which is pretrained on ECHR cases. Since our dataset is relatively small, we do not perform further training of the model on interpretations-specific data. Rather, we use the model as an out-of-the-box classifier for legal interpretations. This corresponds to evaluating the zero-shot performance of LLMs and is therefore a fair baseline for comparisons.

4.2 Prompting strategies

We try several prompting approaches to evaluate which lead to best performance on our dataset. The basic evaluation methods consists of zero-shot and 3-shot learning. We extend the analysis by implementing chain-of-thought reasoning (Wei et al 2023; Kojima et al 2023). This is discussed in Sect. 4.2.1. In all prompting cases we supply a general system text consisting of the task description. The zero-shot setting includes only a test case, while in the 3-shot setting we provide 1 example per interpretation type. The general prompting outline can be seen in Fig. 1.

4.2.1 Extension to reasoning

While work such as Brown et al (2020) suggests LLMs are few-shot learners, Wei et al (2023) show that allowing models to reason by employing chain-of-thought prompting strategies leads to improved performance. Moreover, Kojima et al (2023) find that such models are also more than capable zero-shot reasoners. They recommend adding the phrase *"Let's think step by step."* to prompt the model to thoroughly explain its reasoning. This has been shown to improve results on a variety of benchmarks. Blair-Stanek et al (2023) show this procedure can, in certain cases, lead to improvements in GPT-3's statutory reasoning performance on a simplified dataset of the U.S. tax code. Nori et al (2023) report GPT-4 has impressive potential to answer the multiple-choice questions of the United States Medical Licensing Examination (USMLE). They implement the zero-shot reasoning procedure outlined in Kojima et al (2023). However, they report that basic chain-of-thought prompting does not lead to performance benefits. Liévin et al (2023) evaluate additional domain-specific

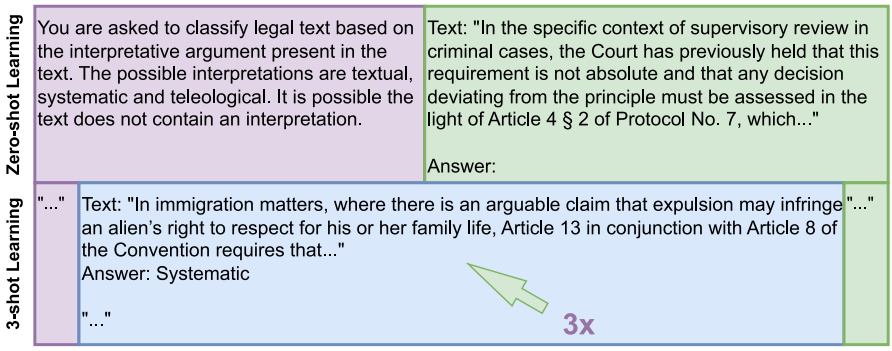


Fig. 1 Prompting template for Zero- and 3-shot learning

chain-of-thought prompts on the most popular medical benchmarks. They report significant variations in performance depending on the provided prompt.

We follow the approach outlined in Kojima et al (2023). We add the phrase *"Let's think step by step."* to the prompts used in the 0-shot learning setting. Figure 2 shows the specific prompt used.

In the 3-shot reasoning case, instead of using the approach outlined above, we provide custom reasoning examples written using expert knowledge. These examples should in a sense guide the models on how to reason on the test cases they are provided. An example prompt can be seen in Fig. 2. In Sect. 5.1 we compare these model-generated reasonings to human-written ones.

4.2.2 A note on the definitions of the interpretations

In all cases we purposefully avoid providing the definitions of the interpretations to the models. As discussed, clear definitions of even the canonical techniques of interpretation are hard to provide. To avoid inducing biases with our selection of the definitions, we rely on the models' internal knowledge of the definitions. We investigate the internal knowledge across models by doing a similarity analysis of the model-generated definitions.

We prompt each model to provide a succinct definition of the 3 interpretations. Using a pretrained Sentence-BERT model (Reimers and Gurevych 2019), a modification of BERT using siamese and triple network structures, we generate text embeddings of the model-generated definitions. More specifically, we use the *all-distilroberta-v1* model with a maximum sequence length set to 512 word pieces. We then run a community detection algorithm that discriminates based on the cosine-similarity score of the embeddings, a widely used measure of similarity between two vectors (Reimers et al 2019; Yang et al 2018). We set the minimum cluster size

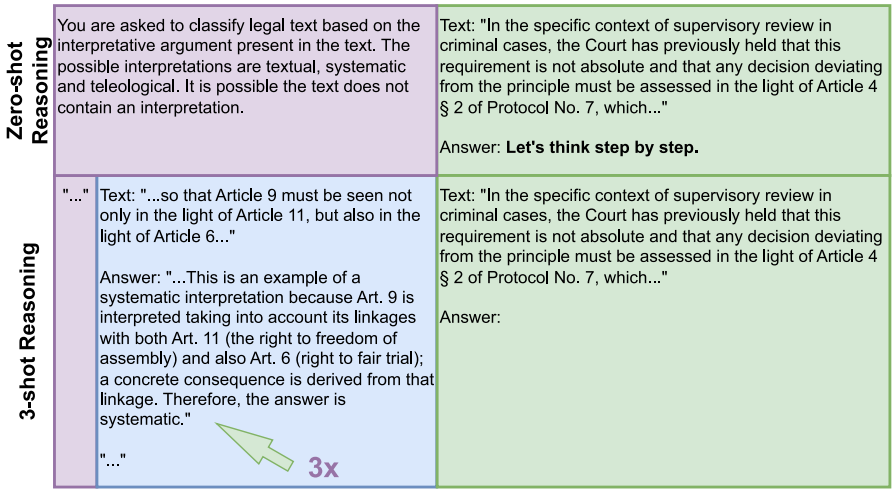


Fig. 2 Prompting template for Zero- and 3-shot reasoning

to 1. The premise is that if the model-generated definitions are similar enough, the community detection algorithm with an appropriate cosine-similarity threshold will separate the definitions into 3 clusters, one for each definition.

We find that with a cosine-similarity threshold of 0.85, all definitions are correctly clustered into 3 clusters with each cluster only containing the appropriate definition. With a threshold of 0.90, all teleological and textual interpretation definitions are contained within the same clusters, while the cluster containing the definitions of a systematic interpretation is missing the definition generated by Mixtral (Fig. 3).

These results imply that the internal knowledge of legal interpretations is very similar across models. This provides further justification for omitting the definitions of the interpretations from the employed prompts and establishes that all models have, to a certain degree, the same base knowledge of legal interpretation techniques.

4.3 Self-consistency

Ye and Durrett (2022) show that in certain cases chain-of-thought prompting might degrade performance compared to standard prompting methods. One possible way of addressing such behaviour is self-consistency as proposed by Wang et al (2023). Self-consistency is the process of sampling a diverse set of reasoning paths and then determining the final answer by finding the most consistent answer in the answer set. They find striking improvements in performance on a variety of benchmarks while also outperforming existing methods for improving language model generation quality. Self-consistency has been used to improve performance of LLMs on medical benchmarks (Singhal et al 2023), and has been further extended with Tree-of-Thoughts and Graph-of-Thoughts frameworks (Long 2023; Yao et al 2023; Besta et al 2024).

We implement self-consistency in the zero-shot reasoning case combined with the *"Let's think step by step"* prompt. In the 3-shot case we follow a similar implementation approach, however, we include custom reasoning examples. We run all

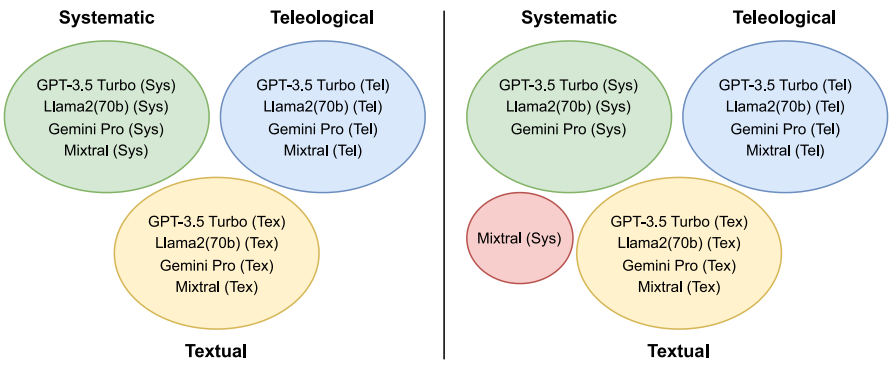


Fig. 3 Clusters after running community detection (**Left:** cosine-similarity threshold equal to 0.85, **Right:** cosine-similarity threshold equal to 0.90; **Sys**=Systematic, **Tel**=Teleological, **Tex**=Textual)

models with temperature set to 1 in order to generate more varied responses and to take advantage of self-consistency.

5 Results

In this section we report the results from the analysis described in Sect. 4. All analysis settings were run for a total of 10 times to obtain more confident performance metrics.

Table 1 shows the increase in accuracy scores moving from zero-shot and 3-shot learning to reasoning, as discussed in Sect. 4.2.1. In other words, we report the change in accuracy when implementing chain-of-thought prompting compared to standard prompting methods. We observe noticeable improvements in the majority of cases, with the most drastic increases found for 3-shot LLama2(70b) and zero-shot GPT−3.5 Turbo whose accuracy scores increased by 37.50% and 39.47%, respectively. However, we also notice a drastic decrease in performance for Gemini Pro in the 3-shot setting. In Sect. 5.1, we show that this decrease in performance can be somewhat mitigated using self-consistency.

We notice rather large increases in 5 of the 8 analysis settings presented in Table 1, with decreasing performance in only 2 cases. This is in line with established research stating that chain-of-thought prompting leads to improvements in most evaluation cases. As discussed in Sect. 4.1, we use LEGAL-BERT as a baseline comparison. We find that LEGAL-BERT achieves an accuracy of 0.32. This is lower than the accuracy achieved by any of the LLMs considered. We investigate the statistical significance of the difference in accuracy scores between our baseline LEGAL-BERT model and each LLM. To this extent, we employ a z-test of two proportions for the null hypothesis $H_0 : \hat{a}_{LB} = \hat{a}_{LLM}$, where $\hat{a}_{LB}$ denotes the accuracy of the LEGAL-BERT model, and $\hat{a}_{LLM}$ denotes the accuracy of an LLM model, at a significance level $\alpha \in (0, 1)$. A rejection of H_0 in this setting is equivalent to saying that, with $(1 - \alpha)\%$ confidence, the LLM classifier is more accurate than the LEGAL-BERT classifier.

Table 1 Comparison of model accuracy between Zero/3-shot learning and reasoning (parentheses denote statistically significant result at a significance level $\alpha = 0.05$)

	Zero-shot			3-shot		
Model	Learning	Reasoning	%-increase	Learning	Reasoning	%-increase
GPT−3.5 Turbo	0.38	(0.53)	39.47	(0.48)	(0.59)	22.92
Geminis Pro	(0.58)	(0.58)	0.0	(0.52)	0.41	−21.15
Llama2(70b)	(0.48)	(0.49)	2.08	0.40	(0.55)	37.50
Mixtral	(0.56)	(0.55)	−1.79	(0.53)	(0.58)	9.43

Table 2 Comparison of model accuracy using Zero-/3-shot reasoning with and without consistency (parentheses denote statistically significant result at a significance level $\alpha = 0.05$)

Model	Zero-shot Reasoning			3-shot Reasoning		
	No Cons	Cons	%-increase	No Cons	Cons	%-increase
GPT−3.5 Turbo	(0.53)	(0.62)	16.98	(0.59)	(0.55)	−6.78
Gemini Pro	(0.58)	(0.65)	12.07	0.41	0.47	14.63
Llama2(70b)	(0.49)	(0.47)	−4.08	(0.55)	(0.54)	−1.82
Mixtral	(0.55)	(0.61)	10.91	(0.58)	(0.62)	6.90

5.1 Effects of self-consistency

Here we report the results when applying the method of self-consistency, described in Sect. 4.3. As discussed previously, chain-of-prompting can lead to degraded performance in certain cases. We have shown this is the case for some models in Table 1.

We see in Table 2 that self-consistency leads to further improvements in performance in most cases, the exception being Llama2(70b) and 3-shot GPT−3.5 Turbo. More importantly, we notice self-consistency leads to an increase in performance in models for which chain-of-thought prompting led to decreased performance, such as Mixtral and Gemini Pro. As stated in the previous section, Gemini Pro saw a massive decrease in performance with 3-shot chain-of-thought prompting. Implementing self-consistency in this situation led to a 14.63% increase in accuracy. However, this was still not enough to surpass the 3-shot learning performance. As another example, zero-shot chain-of-thought prompting led to a minor decrease in performance for Mixtral. However, as we note from the results, self-consistency led to an 10.91% increase in performance, surpassing the original zero-shot learning score. Once again, in terms of accuracy, all LLMs outperform the baseline LEGAL-BERT model.

In Tables 3 and 4 we report F1 scores for the zero-shot and 3-shot settings, respectively. We report individual F1 scores for each label category in a one-vs-rest fashion, as well as Micro F1 and Macro F1 scores. We note from these tables that, across all models, the predictive power for teleological interpretations is most prominent.

Furthermore, we note that, in the zero-shot setting, only GPT−3.5 Turbo and Gemini Pro have any predictive power for the case when there is no interpretation present in the text. Being able to identify a lack of an interpretation is an important aspect of deploying these models for automated legal interpretation data collection, hence models with predictive power in this category are of great importance. Finally, in Table 5 we present the F1 scores for the LEGAL-BERT baseline model. We see the scores are significantly lower than for the LLMs considered in Table 3 and Table 4. In particular, the baseline model has noticeable predictive power only for teleological interpretations. This improved performance on teleological interpretations is in line with what we observe in LLMs.

Table 3 F1 Scores: 0-shot (**top down**: learning, reasoning, self-consistency)

Interpretation	GPT−3.5 Turbo	Llama2(70b)	Gemini Pro	Mixtral
Systematic	0.22	0.45	0.45	0.74
Teleological	0.52	0.64	0.71	0.94
Textual	0.16	0.39	0.64	0.43
No interpretation	0.0	0.0	0.09	0.0
Micro F1	0.40	0.53	0.62	0.74
Macro F1	0.30	0.49	0.60	0.70
Systematic	0.59	0.55	0.56	0.50
Teleological	0.81	0.58	0.76	0.74
Textual	0.30	0.35	0.63	0.45
No interpretation	0.07	0.0	0.08	0.0
Micro F1	0.61	0.51	0.65	0.58
Macro F1	0.56	0.49	0.65	0.56
Systematic	0.64	0.54	0.60	0.59
Teleological	0.84	0.62	0.82	0.75
Textual	0.35	0.29	0.67	0.44
No interpretation	0.01	0.0	0.10	0.0
Micro F1	0.65	0.51	0.71	0.61
Macro F1	0.61	0.48	0.70	0.59

The results show there is still work to be done in developing these models before they can be confidently deployed for collecting legal interpretations data. The performance is quite sensitive depending on the model used and the prompt setting. Hence, converging to the "one best model" is not quite possible. However, what we do observe is possible ways of ensuring the convergence of performance on the given task of classifying legal interpretations. We show that on average, and across models, by using chain-of-thought prompting in combination with self-consistency, we can expect improvements in performance. This allows for procedures where, for example, one performs chain-of-thought prompting and self-consistency on multiple models and then performs majority voting between the model-generated results to obtain an answer closer to ground-truth.

5.2 Analysis of reasoning language

In the 3-shot reasoning setting we provided the models with human-written reasoning examples. We wish to compare how close the models get to emulating human reasoning on interpretative arguments, as well as how the reasoning differs across consistency runs. For this purpose we make use of the Bilingual Evaluation Understudy (BLEU) score (Papineni et al 2002). This metric has been widely used as means of quantifying the quality of machine-translated text compared to human-written references (Thompson et al 2022; Yang et al 2023; Fan et al 2021; Bubeck et al 2023). We use SacreBLEU as introduced by Post (2018) which implements the

Table 4 F1 Scores: 3-shot (**top down**: learning, reasoning, self-consistency)

Interpretation	GPT−3.5 Turbo	Llama2(70b)	Gemini Pro	Mixtral
Systematic	0.13	0.39	0.41	0.37
Teleological	0.70	0.53	0.79	0.86
Textual	0.50	0.05	0.51	0.50
No interpretation	0.17	0.17	0.22	0.45
Micro F1	0.50	0.41	0.58	0.60
Macro F1	0.44	0.32	0.57	0.58
Systematic	0.62	0.66	0.47	0.56
Teleological	0.72	0.65	0.77	0.82
Textual	0.46	0.23	0.39	0.50
No interpretation	0.27	0.33	0.20	0.21
Micro F1	0.62	0.57	0.56	0.65
Macro F1	0.60	0.51	0.54	0.63
Systematic	0.60	0.59	0.47	0.59
Teleological	0.72	0.69	0.81	0.85
Textual	0.38	0.26	0.33	0.48
No interpretation	0.22	0.12	0.20	0.22
Micro F1	0.60	0.57	0.56	0.66
Macro F1	0.57	0.52	0.54	0.64

original BLEU procedure and ensures comparability and reproducibility of BLEU scores by avoiding variations due to different tokenization procedures.

In Table 6 we present the BLEU scores for the 3-shot reasoning case. Furthermore, we investigate similarity in the case of self-consistency. The table includes BLEU scores computed on the entirety of the test set, as well as scores computed on a subset of the data with only correctly labeled samples. We see that the ability to emulate human provided reasoning on legal interpretative techniques is poor across all models, with Llama2(70b) and Mixtral performing convincingly poorer than their counterparts. Gemini Pro, on the other hand, shows the highest capabilities in emulating human-written reasoning.

We do note that the set of human-written reasonings was quite small which could have led to smaller than hoped for BLEU scores since it constrained the set of potential matches between machine-generated reasonings and the human-written ones.

Table 5 F1 Scores: LEGAL-BERT

Interpretation	LEGAL-BERT
Systematic	0.05
Teleological	0.48
Textual	0.0
No interpretation	0.0

Table 6 Comparison of BLEU scores for 3-shot reasoning (**top down:** entire test set, subset of only correctly labeled samples)

	Systematic		Teleological		Textual	
Model	No Cons	Cons	No Cons	Cons	No Cons	Cons
GPT−3.5 Turbo	26.93	27.59	52.68	51.60	20.96	20.21
Gemini Pro	20.19	20.92	54.83	55.97	14.10	13.59
Llama2(70b)	12.28	13.84	17.24	20.01	10.32	10.33
Mixtral	15.45	16.17	18.96	18.95	11.84	11.67
GPT−3.5 Turbo	35.83	34.22	57.53	55.61	35.07	32.19
Gemini Pro	52.26	51.70	75.34	73.51	43.64	43.57
Llama2(70b)	13.84	15.30	17.99	21.82	24.56	21.49
Mixtral	22.07	22.78	19.39	19.30	15.56	15.65

6 Conclusion

The inner functioning of courts is often viewed as opaque from the outside. In this paper, we shed some light on the identifiability of judicial interpretation techniques used by the ECtHR. Research on judicial interpretation of the law is research on the inner functioning of courts because it reveals how the court justifies the outcomes of its decision-making. The use of the techniques of interpretation is, therefore, related to the quality of decision-making by the courts. In the civil law tradition, courts commonly employ a canon of techniques of interpretation (comprising, among others, textual interpretation). The study of judicial interpretation is mostly carried out by lawyers and legal theorists using qualitative methods.

In this paper, we looked at the identifiability of the textual, systematic and teleological interpretation techniques as they are employed by the ECtHR from the perspective of computational law. Despite the relevance of legal interpretation, studies on judicial interpretation from the perspective of computational law are still scarce. A reason is that the use of interpretation techniques is hard to quantify because of the lack of clear-cut definitions of the various techniques and some uncertainty involved in the labelling of individual phrases. In other words, features concerning the techniques of legal interpretation are a complex matter to extract and to classify.

We show how the problem of classification of techniques of legal interpretation can be approached using LLMs. Our experiments revealed, first, that the models used here performed best on the technique of teleological interpretation - i.e. focusing on the purpose and objectives of a legal norm. Second, we demonstrate that LLMs outperform a LEGAL-BERT baseline model on this task. Third, our study is a proof-of-concept on how to automate the classification of the techniques of interpretation used by the ECtHR. This is a task that - if carried out by humans - is very resources- and time-intensive. Fourth, by concentrating on the techniques of legal interpretation, we focus on a "complex" feature of case-law in contrast to features that are easier to extract (e.g. the composition of the judicial panel, the date of the judgment). It is precisely the more complex features of cases that require attention

from computational law because they contain information crucial from a legal perspective.

Our study has a few limitations. A first limitation, from a legal perspective, is that our data only comprises the majority opinion of judgments from 2006-2011. Thus, we cannot exploit data from separate opinions, from decisions or from a larger time range. Secondly, from a computational perspective, LLMs have been referred to as somewhat of a black-box. As discussed, their interpretability is an ongoing discussion. Furthermore, given their nature, the replicability of results also comes into question, especially in settings where obtaining varied responses is key to improving performance. Thirdly, our evaluation was done on relatively short textual excerpts. This is partly due to the difficulty in obtaining labeled data on legal interpretations, but also due to the limits of present-day models. With the largest context window being at 128k tokens, employing these models on large-scale legal documents is not feasible. However, with the announcement of models with much larger context windows, we hope this hurdle will become surmountable. Finally, from a resource-oriented perspective, we do not evaluate the latest GPT-4/GPT-4 Turbo models which reportedly surpass all existing models on a variety of tasks. As discussed, this limitation comes from the sizeable gaps in pricing between those models and the ones evaluated here. This gap in required resources severely affects the feasibility of deploying these models on legal documents at a large scale. Our hope is that with the advancements in model development and increased computational availability, this resource-requirement gap will narrow.

With the development of new models, future work should focus on deploying them on larger legal documents. Furthermore, we show how to gainfully employ LLMs in the process of collecting information on complex legal variables. This is a way to make computational law more relevant to legal research in general. Another aspect we investigated was the models' ability in emulating human-written reasonings. However, as mentioned the set of human-written examples was quite small. Future analyses could use a larger reference set of human-written reasonings in order to investigate if reasoning similarity improves. Finally, another point of view could be taken by using LLMs for generating synthetic data on legal interpretations and then using such data as training samples in more classical machine learning methods.

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